Hello!

# Lecture 6 - The Derivative as a Function

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Homework note:
for Active Calculus & Apex Calculus, problems from exercises section @ bottom of page.

# Quiz Time!

Please get into organized rows so that you have a “bubble” of space around you away from other people.
Reminder:

1. Do not start the quiz until I announce you can start.
2. Do not read off other students’ work to cheat
3. Place all class material (notes, worksheets, etc) and devices (phones, tablets, headphones, etc) away
4. Don’t cheat and or be shady... obviously...

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3. Saw that this quantity was also:
   (1) The slope of the best linear approximation

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   (2) The slope of the tangent line to the graph of $y = f(x)$ at $(a, f(a))$.
   (3) The instantaneous velocity of a particle whose motion is described by $f(x)$ at $x = a$.

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\begin{aligned}
f'(0) &= \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \qquad (3) \\
&= \lim_{x \to 0} \frac{x^2}{x}
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Wait... these all seem pretty similar!

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For a function $f(x)$

Is this different from what we were doing before?

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$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{u \to x} \frac{f(u) - f(x)}{u - x} \tag{12}$$

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$f'(x) = 2x$

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So we immediately get that:

1. $f'(0) = 2 \cdot 0 = 0$
2. $f'(1) = 2 \cdot 1 = 2$
3. $f'(2) = 2 \cdot 2 = 4$

# The Derivative Function III

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   (1) For which values is $f'(a)$ positive? (when is $f(x)$ increasing?)



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2. Information Extraction. Without the equation of the function, it's *impossible* to analytically study interesting properties of the derivative like:
   (1) For which values is $f'(a)$ positive? (when is $f(x)$ increasing?)
   (2) For which values is $f'(a)$ zero? (important for maximizing/minimizing which we'll do in 3/4 weeks)

$x^2 \rightsquigarrow$ factoring or roots

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## Big Idea of Making it a Function

Here're two very important reasons why we might want to treat the derivative as a function and not just compute value by value:

1. Speed. Once we do the (slightly length) work of computing the general functional form of the derivative all future derivatives at a point can be computed *so quickly!* just by evaluation!
2. Information Extraction. Without the equation of the function, it's *impossible* to analytically study interesting properties of the derivative like:
   (1) For which values is $f'(a)$ positive? (when is $f(x)$ increasing?)
   (2) For which values is $f'(a)$ zero? (important for maximizing/minimizing which we'll do in 3/4 weeks)
   (3) For which values is $f'(a)$ undefined? (when can we not linearly approximate?)

$x^2 \rightsquigarrow$ factoring or roots

$\frac{1}{x-1} \rightsquigarrow x = 1$

# The Derivative Function III

## Big Idea of Making it a Function

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   (2) For which values is $f'(a)$ zero? (important for maximizing/minimizing which we'll do in 3/4 weeks)
   (3) For which values is $f'(a)$ undefined? (when can we not linearly approximate?)

   but given an equation of a function it's fairly easy to answer all of those!

# The Derivative Function IV

For example: at which $x$ values is the slope of the line tangent to $f(x) = x^3 - 2x + 1$ equal to zero?

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# The Derivative Function IV

For example: at which $x$ values is the slope of the line tangent to $f(x) = x^3 - 2x + 1$ equal to zero?

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# The Derivative Function IV

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Step 2: Compute derivative:

# The Derivative Function IV

For example: at which $x$ values is the slope of the line tangent to $f(x) = x^3 - 2x + 1$ equal to zero?

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Step 2: Compute derivative:

$$f'(x) =$$

# The Derivative Function IV

For example: at which $x$ values is the slope of the line tangent to $f(x) = x^3 - 2x + 1$ equal to zero?

Step 1: Translate to derivative question: when is $f'(x) = 0$?

Step 2: Compute derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

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Step 2: Compute derivative:

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{(x+h)^3 - 2(x+h) + 1 - (x^3 - 2x + 1)}{h} \end{aligned} \tag{18}$$

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$$= \lim_{h \to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h + 1 - x^3 + 2x - 1}{h}$$

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$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - 2h}{h}$$

$= 3x^2 + 3xh + h^2 - 2$

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# The Derivative Function V

Last slide: $f'(x) = 3x^2 - 2$.

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Last slide: $f'(x) = 3x^2 - 2$.
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$$3x^2 - 2 = 0 \tag{24}$$
$$3x^2 = 2$$

# The Derivative Function V

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$$3x^2 - 2 = 0 \tag{24}$$
$$3x^2 = 2 \tag{25}$$
$$x^2 = \frac{2}{3}$$

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Step 4: Translate derivative question answer to original question answer:

# The Derivative Function V

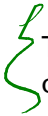
Last slide: $f'(x) = 3x^2 - 2$.
Step 3: Answer derivative question: when is $f'(x) = 3x^2 - 2 = 0$?

$$3x^2 - 2 = 0 \tag{24}$$
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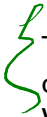
When $x$ is $\pm\sqrt{\frac{2}{3}}$, the slope of the line tangent to $f(x) = x^3 - 2x + 1$ is equal to zero.

# The Derivative Function VI



This is the general strategy that you can use to solve *sooo* many questions about the behavior of functions!

# The Derivative Function VI



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We'll see that in the worksheet for today :)

# The Derivative Function VII

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1. For some random function $f(x)$ whose *domain* is all real numbers $\mathbb{R}$ must it also be the case that the domain of $f'(x)$ is also $\mathbb{R}$?

1) If $f(x)$ was not cont. $x=a$ then $f'(a)$ was not defined

cont.

# The Derivative Function VII

Some conceptual questions:

1. For some random function $f(x)$ whose *domain* is all real numbers $\mathbb{R}$ must it also be the case that the domain of $f'(x)$ is also $\mathbb{R}$?
2. How do the following conditions relate:

# The Derivative Function VII

Some conceptual questions:

1. For some random function $f(x)$ whose *domain* is all real numbers $\mathbb{R}$ must it also be the case that the domain of $f'(x)$ is also $\mathbb{R}$?
2. How do the following conditions relate:

(1) $f(x)$ is defined at $x = a$
(2) $\lim_{x \to a} f(x)$ exists
(3) $f(a) = \lim_{x \to a} f(x)$
(4) The domain of $f(x)$ is all real numbers $\mathbb{R}$
(5) $\lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}$ exists
(6) $f'(a)$ exists
(7) $f'(x)$ is defined at $x = a$
(8) $f(x)$ is differentiable at $x = a$
(9) $f'(x)$ is continuous at $x = a$
(10) $f'(x)$ is differentiable at $x = a$
(11) The domain of $f'(x)$ is all real numbers $\mathbb{R}$



# The Derivative Function VII

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(7) $f'(x)$ is defined at $x = a$
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(9) $f'(x)$ is continuous at $x = a$
(10) $f'(x)$ is differentiable at $x = a$
(11) The domain of $f'(x)$ is all real numbers $\mathbb{R}$

3. Which of the above conditions relied on $f'(x)$ *being a function* rather than being a limit defined at a point $x = a$?

# Worksheet Time!

Let’s get into groups and work on the worksheet!

# End of Class!

That’s all folks :)